

| Question |     |      |    | Marking details                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|-----|------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)  |    | one {cell/ bacteria} forms one colony<br>Accept there is no clumping/ or description of                                                                                                                                            | 1               |     |     | 1     |       | 1    |
|          |     | (ii) | I  | $3.1 \times 10^5 = 3$ marks<br><br>Award 2 marks for 310000<br><br>Award 1 mark for 31 colonies<br>Error in counting – 30 or 32<br><br>Award 2 marks for $3.2 \times 10^5$<br>$3 \times 10^5$<br><br>Award 1 mark 320000 or 300000 |                 | 3   |     | 3     | 3     | 3    |
|          |     |      | II | Correct conclusion based on answer calculated in I.<br>Eg. No, it was not safe as the number of bacteria was above the {safe limit/ $1 \times 10^5$ }                                                                              |                 |     | 1   | 1     |       | 1    |
|          | (b) |      |    | To test for the presence of human pathogens/ OWTTE<br>Ignore reference to temperature of body                                                                                                                                      |                 |     | 1   | 1     |       | 1    |
|          | (c) | (i)  |    | To prevent {contamination/ or description of} of {environment/ agar plates}                                                                                                                                                        | 1               |     |     | 1     |       | 1    |